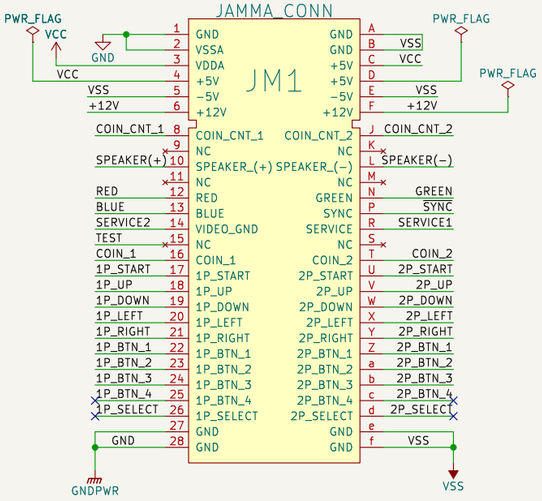
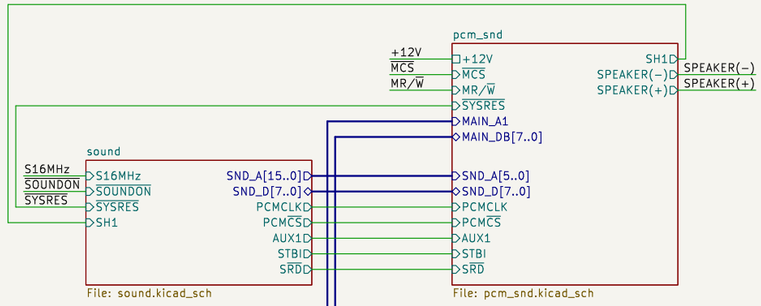
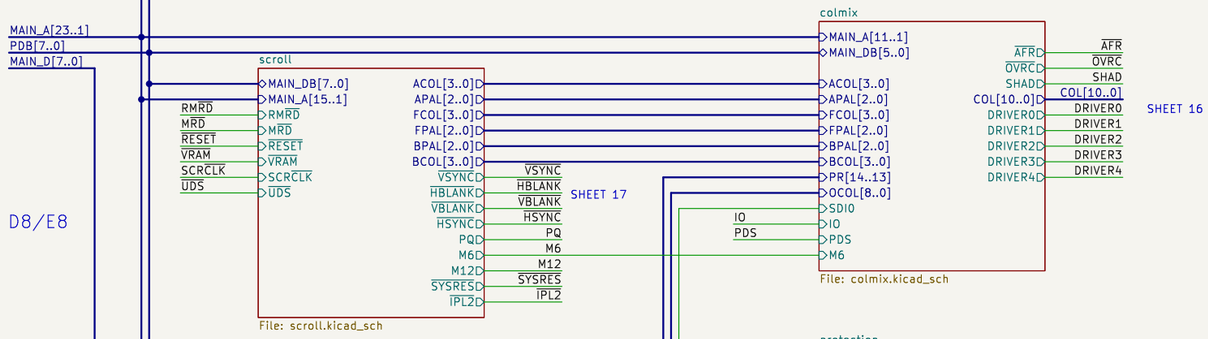
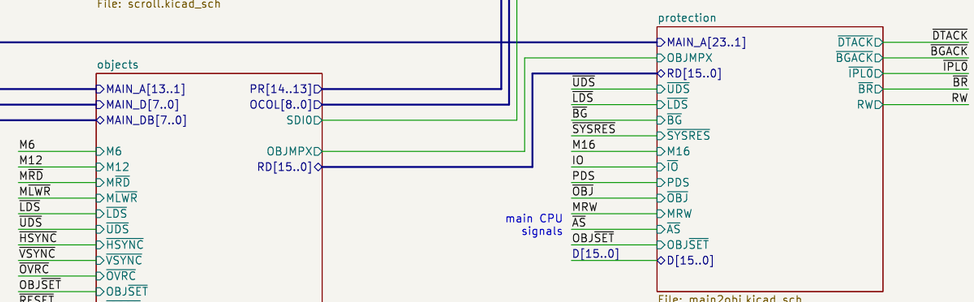








PDB[7:0] is either main CPU upper or lower byte depending on UDS/LDS See sch. page 18, D8/E8

PDS=UDS&LDS

These schematics cover the missing pages in the original ones published by the maker. Signal names existing in the original pages have been preserved. New signal names try to follow the maker style or are taken from similar schematics of the same maker.

JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/riders/sch/ssriders
KONAMI PWB 352359C

Sheet: /
File: ssriders.kicad_sch

Title: Sunset Riders

Size: A3	Date: 2023-08-29	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 1/7

MAIN_A[11..1]

G13

SHEET 16

MAIN_A8 1
MAIN_A9 2
MAIN_A10 3
PDSD 6
IOD 5
MAIN_A11 4

A0 A1 A2 address decoder E3 E2 E1

O0 O1 O2 O3 O4 O5 O6 O7

15 DRIVER0 PIN1 B13
14 DRIVER1 PIN15 C18
13 DRIVER2 PIN11 C15
12 DRIVER3 PIN9 C14
11 AFR
10 OVRC
9 DRIVER4 PIN5 F18
7 PCUCS

74LS138

On the schematic it appears as 17F,
but on the PCB it is 18F.

MAIN_DB[5..0]

PR[14..13]

VCC

VSS

6M M6

CSDIO

MAIN_DB0 MAIN_DB1 MAIN_DB2 MAIN_DB3 MAIN_DB4 MAIN_DB5 MAIN_A1 MAIN_A2 MAIN_A3 MAIN_A4

PR14 PR13

PRIORITY

COLOR INPUTS

CHIP SELECT INPUT
COLOR OUTPUTS

J4
KONAMI

053251
PRIORITY CONTROL UNIT

30 pins top/bottom
20 pins left/right

PCUCS COL0 COL1 COL2 COL3 COL4 COL5 COL6 COL7 COL8 COL9 COL10

COL[10..0]

SDO0 SHAD
BRIT BRIT pdf page 17

OCOL[8..0]

OCOL0 OCOL1 OCOL2 OCOL3 OCOL4 OCOL5 OCOL6 OCOL7 OCOL8

FCOL0 FCOL1 FCOL2 FCOL3 FPAL0 FPAL1 FPAL2

FCOL[3..0]
FPAL[2..0]

ACOL0 ACOL1 ACOL2 ACOL3 APAL0 APAL1 APAL2

ACOL[3..0]
APAL[2..0]

BCOL0 BCOL1 BCOL2 BCOL3 BPAL0 BPAL1 BPAL2

BCOL[3..0]
BPAL[2..0]

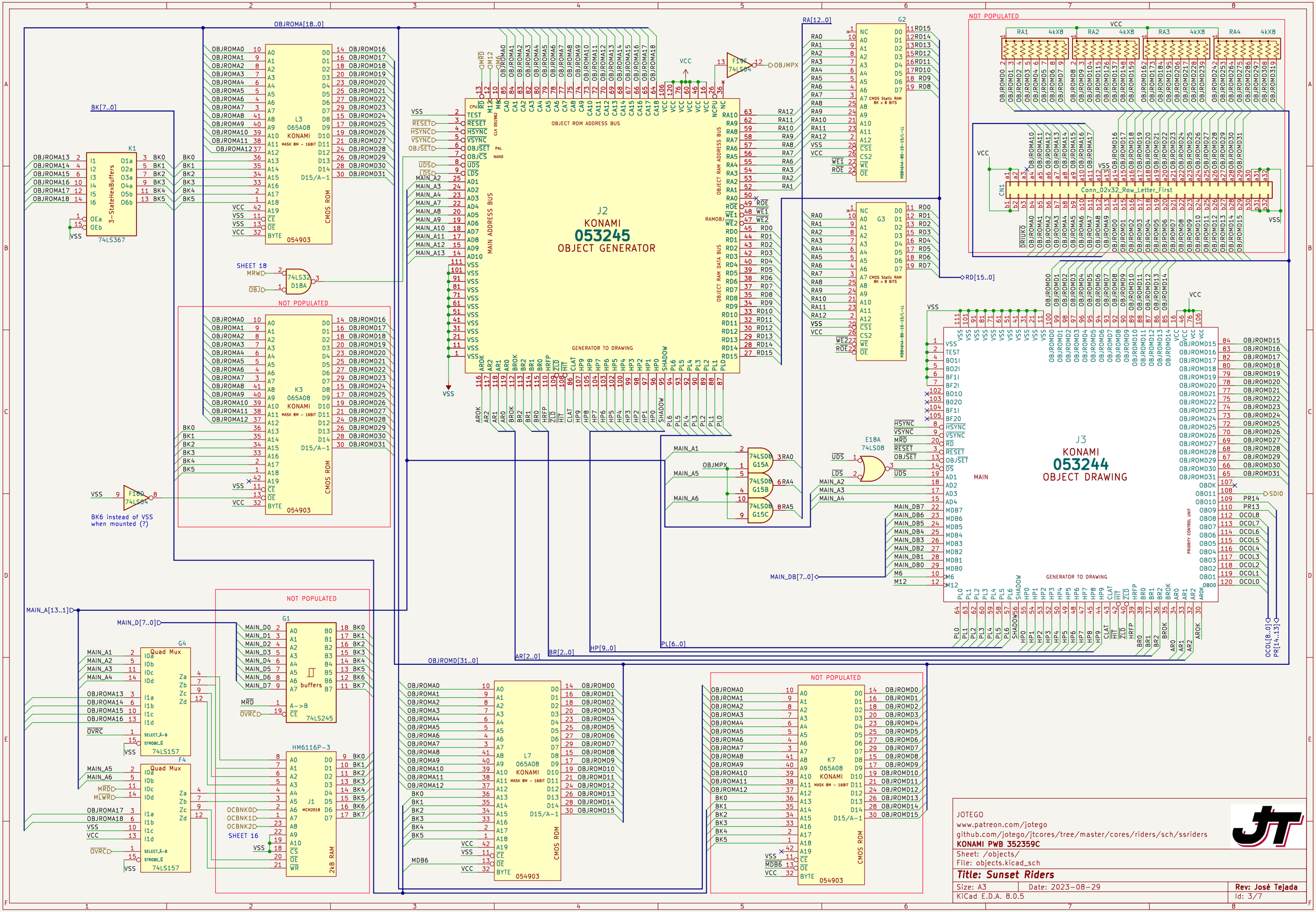
SHEET 02

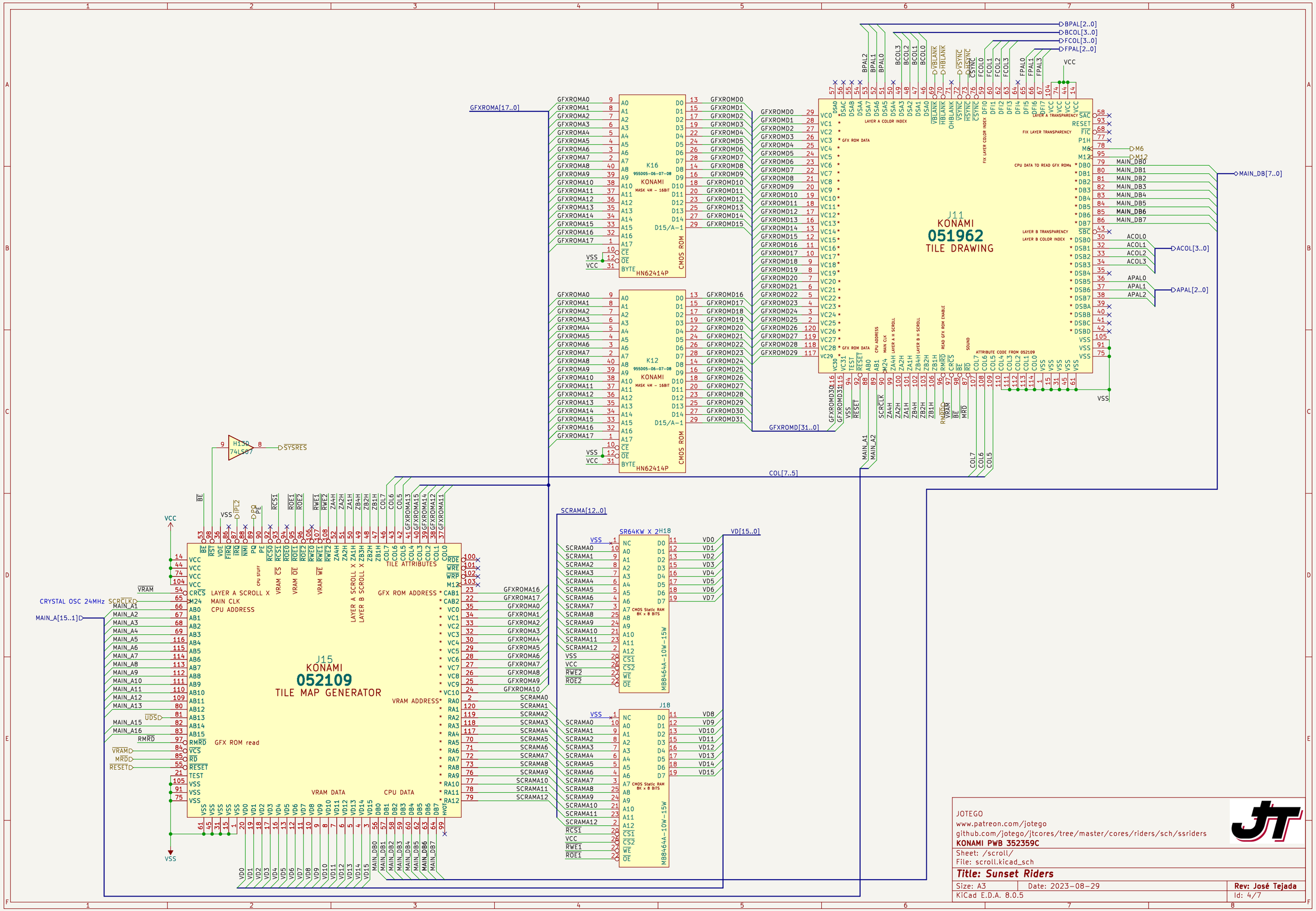
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github.com/jotego/jtcores/tree/master/cores/riders/sch/ssriders
KONAMI PWB 352359C

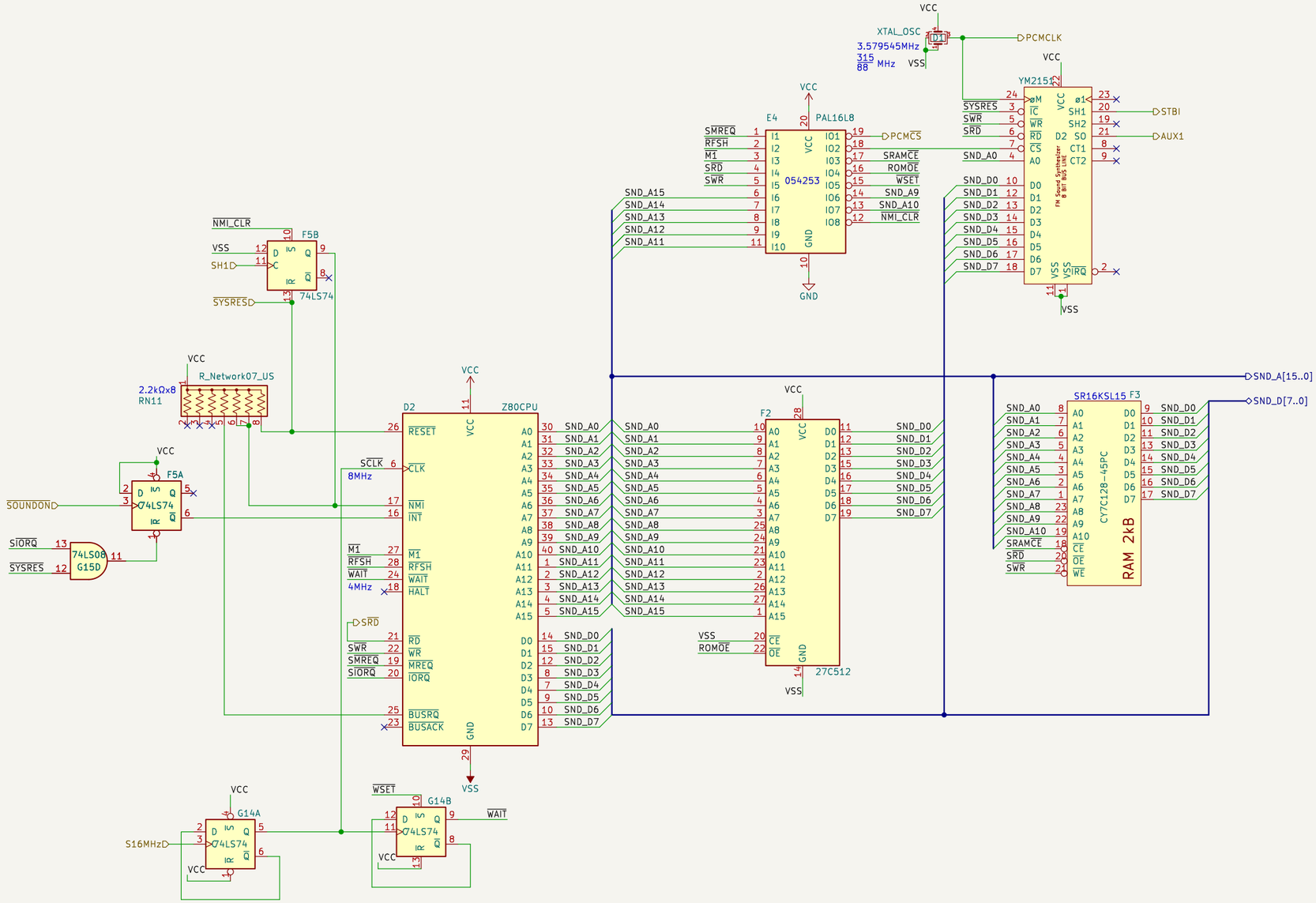
Sheet: /colmix/
File: colmix.kicad_sch

Title: Sunset Riders

Size: A3	Date: 2023-08-29	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 2/7


OBJROMA[18..0]
BK[7..0]
K1
74LS367
3-StateHexBuffers
L3
065A08
KONAMI
MASK 8M - 16BIT
CMOS ROM
054903
SHEET 18
74LS32
D18A
NOT POPULATED
K3
F18D
74LS04
BK6 instead of VSS
when mounted (?)
J2
KONAMI
053245
OBJECT GENERATOR
OBJECT ROM ADDRESS BUS
MAIN ADDRESS BUS
OBJECT RAM ADDRESS BUS
OBJECT RAM DATA BUS
GENERATOR TO DRAWING
RAMOBJ
F19F
74LS04
OBJMPX
RA[12..0]
G2
G3
HM6864A-80-10-15/L-LL
CMOS Static RAM
8K x 8 BITS
RD[15..0]
NOT POPULATED
RA1 4kX8
RA2 4kX8
RA3 4kX8
RA4 4kX8
CN1
Conn_02x32_Row_Letter_First
DRUK0
J3
KONAMI
053244
OBJECT DRAWING
MAIN
PRIORITY CONTROL UNIT
E18A
74LS08
74LS08 G15A
74LS08 G15B
74LS08 G15C
MAIN_DB[7..0]
SDI0
OCOL[8..0]
PR[14..13]
G1
74LS245
buffers
A->B
G4
F4
Quad Mux
74LS157
SELECT_A-B
STROBE_G
MAIN_A[13..1]
MAIN_D[7..0]
HM6116P-3
2kB RAM
J1
MCM2018
SHEET 16
OCBNK0
OCBNK1
OCBNK2
OBJROMD[31..0]
AR[2..0]
BR[2..0]
HP[9..0]
PL[6..0]
JOTEGO
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github.com/jotego/jtcores/tree/master/cores/riders/sch/ssriders
KONAMI PWB 352359C
Sheet: /objects/
File: objects.kicad_sch
Title: Sunset Riders
Size: A3
Date: 2023-08-29
KiCad E.D.A. 8.0.5
Rev: José Tejada
Id: 3/7


J11
KONAMI
051962
TILE DRAWING
J15
KONAMI
052109
TILE MAP GENERATOR
K16
K12
HN62414P
CMOS ROM
J18
SR64KW X 2H18
MB8464A-10W-15W
CMOS Static RAM 8K x 8 BITS
74LS07
CRYSTAL OSC 24MHz
BPAL[2..0]
BCOL[3..0]
FCOL[3..0]
FPAL[2..0]
MAIN_DB[7..0]
ACOL[3..0]
APAL[2..0]
GFXROMA[17..0]
GFXROMD[31..0]
COL[7..5]
SCRAMA[12..0]
VD[15..0]
MAIN_A[15..1]
LAYER A COLOR INDEX
GFX ROM DATA
FIX LAYER TRANSPARENCY
CPU DATA TO READ GFX ROMs
LAYER B TRANSPARENCY
LAYER B COLOR INDEX
ATTRIBUTE CODE FROM 052109
LAYER A SCROLL X
LAYER A SCROLL Y
LAYER B SCROLL X
LAYER B SCROLL Y
MAIN CLK
CPU ADDRESS
GFX ROM read
VRAM ADDRESS
VRAM DATA
CPU DATA
TILE ATTRIBUTES
GFX ROM ADDRESS
JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/riders/sch/ssriders
KONAMI PWB 352359C
Sheet: /scroll/
File: scroll.kicad_sch
Title: Sunset Riders
Size: A3
Date: 2023-08-29
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 4/7

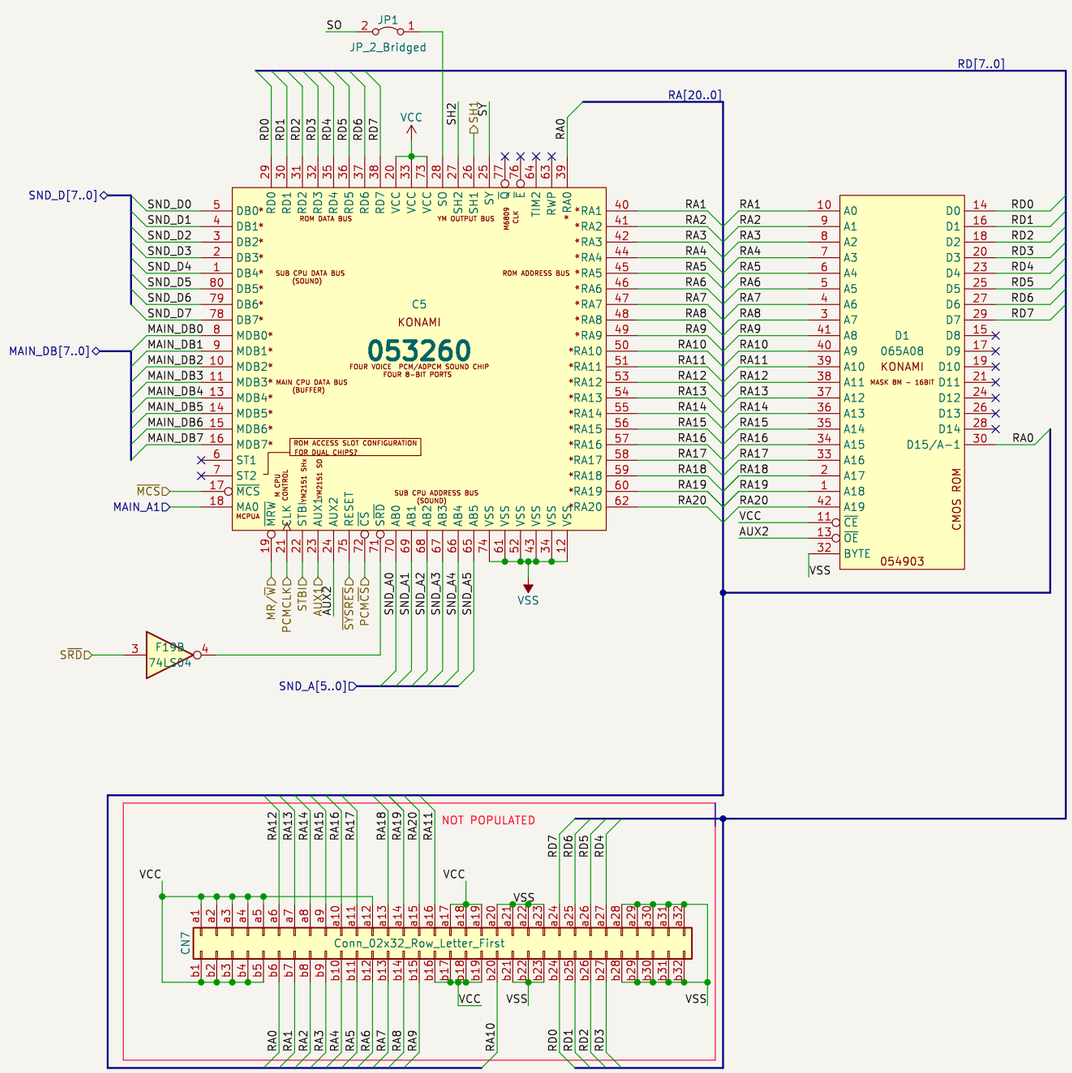


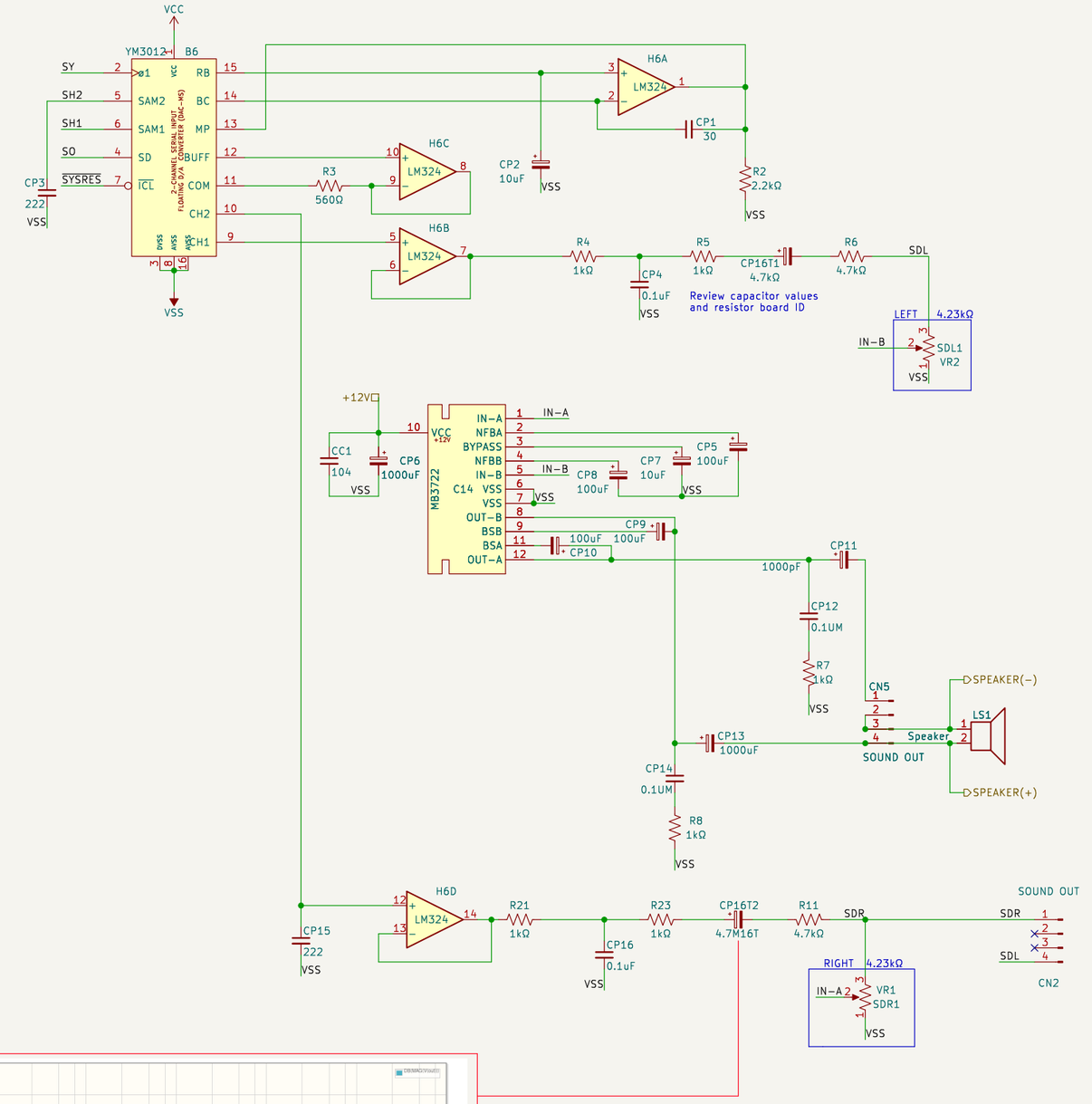
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github.com/jotego/jtcores/tree/master/cores/riders/sch/ssriders
KONAMI PWB 352359C

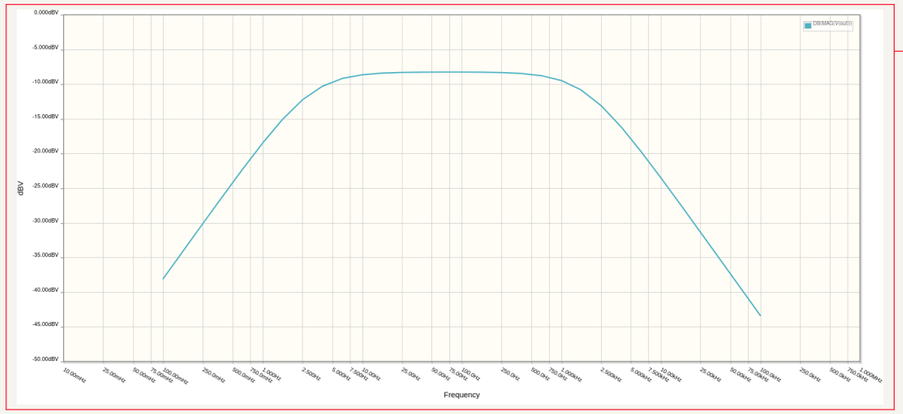
Sheet: /sound/
File: sound.kicad_sch

Title: Sunset Riders

Size: A3	Date: 2023-08-29	**Rev: José Tejada**
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Sheet: /pcm_snd/
File: pcm_snd.kicad_sch

Title: Sunset Riders

Size: A3	Date: 2023-08-29	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 6/7

main CPU data bus
to object RAM

D[15..0]
main CPU
data bus

RD[15..0]

K8
J8
74LS245
buffers
A->B
CE

H7
KONAMI
054358

-DMA access to main CPU 520us
-Returns a number based on
a CPU written RAM value
-updates obj priority data

19 pins top/bottom
13 pins left/right
64 pins total

MAIN_A[23..1]

VCC
VSS
to M68000
R1
1KΩ

F19F
74LS04
D17B
74LS00
E18C
74LS08
D17A
74LS00
D17C
E18D
74LS08

MRW
OBJMPX
OBJSET
OBJ

PAL MAIN
SHEET 18

F5
PAL20RS10
054618

VPAD
OBJMPXD
MDTACK
VRAM
OBJ
OBJSET
COLOR
IO
MBOE
MRAMOEWR
MROMOE2
MROMOE1

F15A
74LS74
PDS
PQ
MAIN_A17

SHEET 09

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github.com/jotego/jtcores/tree/master/cores/riders/sch/ssriders
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Sheet: /protection/
File: main2obj.kicad_sch

Title: Sunset Riders

Size: A3	Date: 2023-08-29	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 7/7